Agentic AI Learning Journey: 30-Day Practical Roadmap

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Introduction to Agentic AI

Welcome to your 30-day roadmap for mastering Agentic AI.
Agentic AI represents a shift from static prompt-response models to autonomous systems that can plan, use tools, and execute multi-step workflows.
This structured guide will take you from foundational Python and LLM API concepts to building and deploying autonomous agents.
Dedicate 2 to 3 hours daily to complete the tasks, review code, and build your portfolio.

Prerequisites & Core Philosophy

Before starting, ensure you have a working Python environment, an OpenAI (or alternative LLM) API key, and basic familiarity with Git.
The roadmap emphasizes hands-on implementation over purely theoretical study.
You will learn by building: prompt chaining, routing, tool usage, planning loops, and multi-agent collaboration frameworks.

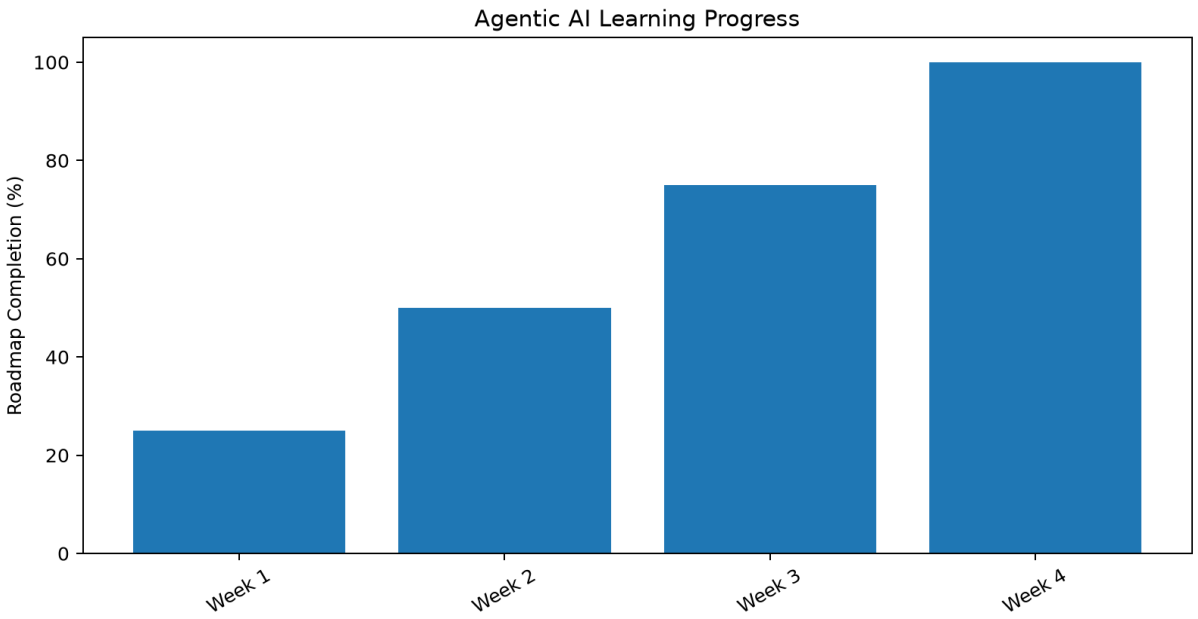
Job Preparation and Next Steps

Upon completing this 30-day plan, you will have built several fully functional agentic workflows.
Use your final days to polish your GitHub repository, document your projects with clear README files, and prepare for technical discussions regarding autonomous agent architectures.
Remember that consistent practice and building real projects are the keys to mastering this rapidly evolving domain.

Day	Tasks
Day 1	Set up Python environment, virtual environments, and install basic libraries (openai, requests, python-dotenv).
Day 2	Review LLM API basics: Chat completions, system prompts, temperature settings, and JSON mode.
Day 3	Learn prompt engineering patterns for agents: Chain of Thought (CoT) and Least-to-Most prompting.
Day 4	Build a basic prompt-chaining script that takes user input, refines it via LLM, and generates a final report.
Day 5	Introduce routing patterns: Classify user intent and direct queries to specialized prompt handlers.
Day 6	Study the concept of agent memory: Short-term (conversation history) and long-term (vector embeddings).
Day 7	Weekly Review & Practice: Refactor Days 1-6 code, add error handling, and push to a GitHub repository.
Day 8	Introduction to Tool Use (Function Calling): Learn how LLMs output structured tool calls.
Day 9	Build a custom calculator and web search tool and bind them to an LLM completion loop.
Day 10	Implement the ReAct (Reason + Act) loop manually from scratch using raw Python loops.
Day 11	Explore framework ecosystems: Overview of LangChain, LangGraph, CrewAI, and AutoGen.
Day 12	Build your first agent using LangChain Expression Language (LCEL) and basic toolkits.

Day	Tasks
Day 13	Implement memory persistence in your agent using SQLite or simple JSON file storage.
Day 14	Weekly Review & Practice: Test edge cases, handle infinite loops, and handle API rate limits gracefully.
Day 15	Introduction to multi-agent systems: Roles, communication protocols, and task delegation.
Day 16	Build a simple two-agent system (e.g., Writer and Editor) using Python classes.
Day 17	Explore CrewAI framework: Define agents, tasks, and sequential processes.
Day 18	Explore AutoGen framework: Setup conversational agents for collaborative problem solving.
Day 19	Implement human-in-the-loop validation checkpoints before critical agent actions.
Day 20	Build a Retrieval-Augmented Generation (RAG) tool for your agent using ChromaDB or FAISS.
Day 21	Weekly Review & Practice: Optimize agent prompts, reduce token usage, and clean up repository.
Day 22	Introduction to Agentic workflows for software development (Planning, Coding, Testing).
Day 23	Build an autonomous code-review agent that analyzes Python scripts and suggests improvements.
Day 24	Explore agent evaluation metrics: Measuring task success rate, latency, and cost.
Day 25	Containerize your agent application using Docker for consistent deployment.
Day 26	Build a simple web interface (Streamlit or Gradio) for your agent application.
Day 27	Deploy your agent application to a cloud provider (e.g., Hugging Face Spaces or Render).
Day 28	Mock Interview & Portfolio Prep: Document your projects, write comprehensive README files, and prepare arch
Day 29	Final code audit, security check (ensuring no hardcoded API keys), and license addition.
Day 30	Graduation & Next Steps: Review achievements, outline advanced topics (like fine-tuning or reinforcement learn

Agentic AI Learning Progress


Agentic AI Learning Progress
Roadmap Completion (%)
0
20
40
60
80
100
Week 1
Week 2
Week 3
Week 4